1.8 Integration

Mathematics — OCR A-Level (H240) — Paper 1: Pure Mathematics

Integration is the reverse process of differentiation. Where differentiation tells you how a quantity is changing, integration lets you recover the quantity from its rate of change — or, geometrically, compute the area under a curve. The two operations are bound together by the *fundamental theorem of calculus*, one of the most important results in mathematics. This chapter develops the techniques of indefinite and definite integration — power rule, standard integrals, substitution, integration by parts, partial fractions, and trigonometric identities — and uses them to find areas, volumes of revolution, and solutions to first-order separable differential equations. Every applied problem involving accumulation, total change, work, or volume ultimately rests on integration.

1. Integration as the Reverse of Differentiation

If differentiation takes a function $F(x)$ and produces its derivative $f(x) = F'(x)$, then integration runs the process backwards: given $f(x)$, find a function whose derivative is $f(x)$. Such a function is called an *antiderivative* of f. The essential subtlety is that antiderivatives are not unique — adding any constant to an antiderivative still yields an antiderivative, because the derivative of a constant is zero.

Indefinite integral (antiderivative). An *antiderivative* of $f(x)$ is any function $F(x)$ with $F'(x) = f(x)$. The set of all antiderivatives is denoted \[\int f(x)\, dx = F(x) + C, \] where C is an arbitrary constant — the *constant of integration*. The symbol $\int$ is an elongated "S" (from "sum"); dx indicates the variable of integration.

Why the constant? If $F'(x) = f(x)$ and $G(x) = F(x) + C$ for any constant C, then $G'(x) = F'(x) + 0 = f(x)$ as well. So F and G are equally valid antiderivatives. Conversely, any two antiderivatives of f differ by a constant (this is a consequence of the mean value theorem). The "$+C$" captures this ambiguity.

Worked Example 1 — Why the constant matters

```
1.  Consider f(x) = 2x.
2.  F(x) = x²        has derivative 2x ✓
3.  G(x) = x² + 7    has derivative 2x ✓
4.  H(x) = x² − 3.4  has derivative 2x ✓
5.  All three are antiderivatives of 2x — they differ only by a constant.
6.  We capture all of them at once by writing:
7.      ∫ 2x dx = x² + C.

// The +C is mandatory in every indefinite integral. Losing it loses a mark.
```

Exam tip — always write +C for indefinite integrals. OCR mark schemes list the constant of integration as a standalone awarded mark on almost every indefinite-integral question. Forgetting it is the single most common error at A-level. The only place you don't need $+C$ is in a *definite* integral, because the constant cancels when you subtract $F(a)$ from $F(b)$.

2. The Power Rule for Integration

Reversing the power rule $\frac{d}{dx}(x^{n+1}) = (n+1)x^n$ gives the integration power rule. Divide by $n+1$ to isolate x^n:

Power rule for integration. For any real $n \neq -1$, \[\int x^n\, dx = \frac{x^{n+1}}{n+1} + C. \] In words: *add one to the power, divide by the new power*. The exclusion $n = -1$ is needed because division by zero would occur. The case $n = -1$ is handled by the logarithm: \[\int \frac{1}{x}\, dx = \ln|x| + C. \] The absolute value is essential — $\ln$ is only defined for positive arguments, but $1/x$ is defined for all $x \neq 0$.

The power rule extends term-by-term via linearity: for constants a, b and integrable functions f, g,

$$\int \big[af(x) + bg(x)\big]\, dx = a\int f(x)\, dx + b\int g(x)\, dx.$$

Worked Example 2 — Polynomial with mixed powers

Find $\int (3x^2 - 4x + 1)\, dx$.

```
1.  Integrate term by term using the power rule:
2.     ∫ 3x² dx = 3 · (x³/3) = x³.
3.     ∫ −4x dx = −4 · (x²/2) = −2x².
4.     ∫ 1 dx  = x           (since 1 = x⁰, add 1 to get x¹/1.)
5.  Combine and add the constant:
6.     ∫ (3x² − 4x + 1) dx = x³ − 2x² + x + C.
7.  Check by differentiating: d/dx (x³ − 2x² + x + C) = 3x² − 4x + 1 ✓.

// Always verify by differentiating — it catches arithmetic errors in seconds.
```

3. Standard Integrals

Reversing the standard derivatives gives a short list of integrals that must be memorised. Each is just a differentiation rule read backwards.

Function $f(x)$	Integral $\int f(x)\, dx$	Notes
$x^n \ (n \neq -1)$	$\dfrac{x^{n+1}}{n+1} + C$	Any real n except -1.
$1/x$	$\ln\lvert x\rvert + C$	Absolute value for $x < 0$.
$\sin x$	$-\cos x + C$	Minus sign — easily missed.
$\cos x$	$\sin x + C$	No minus sign.
$\sec^2 x$	$\tan x + C$	Reverses $(\tan x)' = \sec^2 x$.
e^x	$e^x + C$	Unique: same as function.
e^{kx}	$\dfrac{1}{k} e^{kx} + C$	Divide by the inner coefficient.

$\sin(kx)$	$-\frac{1}{k}\cos(kx)+C$	Chain-rule reversal.
$\cos(kx)$	$\frac{1}{k}\sin(kx)+C$	Chain-rule reversal.
$\frac{1}{ax+b}$	$\frac{1}{a}\ln\lvert ax+b\rvert+C$	Linear inner — divide by a.

Exam tip — watch for the sign on sin and cos. It is easy to mix up which of $\sin$ and $\cos$ picks up a minus sign on integration. The safest rule: *differentiate your answer to check*. For $\int \sin x\,dx = -\cos x + C$: differentiate $-\cos x$ to get $-(-\sin x) = \sin x$ ✓. For $\int \cos x\,dx = \sin x + C$: differentiate $\sin x$ to get $\cos x$ ✓. A five-second check prevents a lost mark.

4. Definite Integrals and the Fundamental Theorem

An indefinite integral produces a family of antiderivatives. A *definite* integral produces a number — the net change of an antiderivative between two specified limits, equivalently the (signed) area between the curve and the x-axis.

Definite integral. If F is an antiderivative of f on $[a,b]$, then the definite integral of f from a to b is \[\int_a^b f(x)\, dx = \Big[F(x)\Big]_a^b = F(b) - F(a). \] The numbers a and b are the lower and upper *limits of integration*. No constant of integration is written — it would cancel in $F(b)+C-(F(a)+C)$.

Fundamental theorem of calculus. If f is continuous on $[a,b]$, then $F(x)=\int_a^x f(t)\,dt$ is differentiable on (a,b) and $F'(x)=f(x)$. Consequently, if F is any antiderivative of f on $[a,b]$, then \[\int_a^b f(x)\, dx = F(b) - F(a). \] This theorem ties the two branches of calculus together: the integral is the inverse of the derivative.

Useful properties of definite integrals:

- **Swap limits, change sign:** $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$.
- **Adjacent interval additivity:** $\int_a^b f\,dx + \int_b^c f\,dx = \int_a^c f\,dx$.
- **Zero interval:** $\int_a^a f(x)\,dx = 0$.

- **Linearity:** $\int_a^b [\alpha f + \beta g]\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g\,dx$.

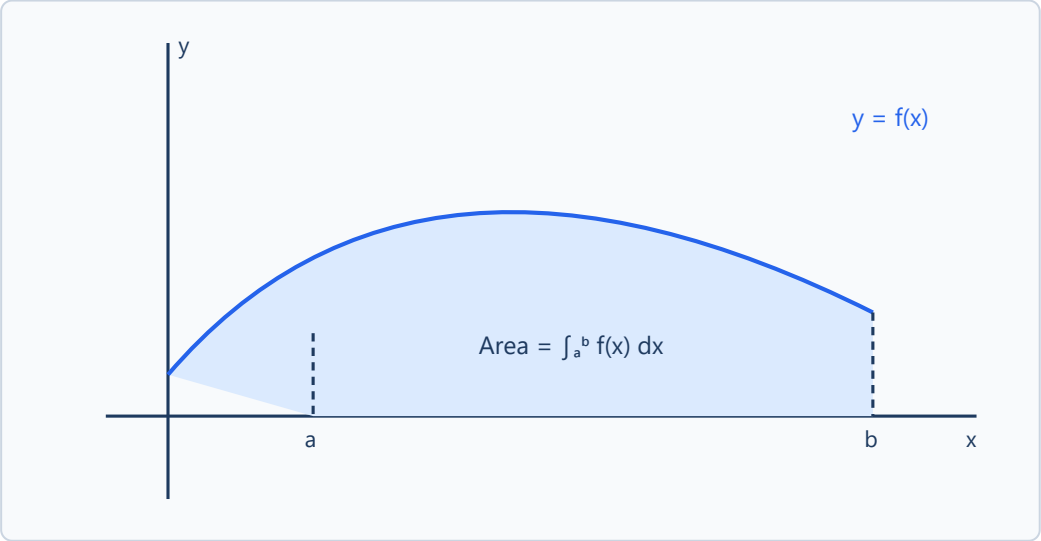
Worked Example 3 — Evaluate $\int_1^3 (2x+1)\,dx$

```
1.  Find an antiderivative:
2.     ∫ (2x + 1) dx = x² + x    (+C, but we may drop C for definite integrals).
3.  Apply the limits:
4.     [x² + x]₁³ = (3² + 3) − (1² + 1)
5.                = (9 + 3)   − (1 + 1)
6.                = 12 − 2
7.                = 10.
8.  ∫₁³ (2x + 1) dx = 10.

// Geometric check: this integral is the area of a trapezium under y = 2x + 1
// from x = 1 to x = 3 — heights 3 and 7, width 2, area = ½(3+7)(2) = 10 ✓.
```

5. Area Under a Curve

If $f(x) \geq 0$ throughout $[a, b]$, the definite integral $\int_a^b f(x)\,dx$ equals the area enclosed between the curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$. If $f(x) < 0$ somewhere on the interval, the integral measures a *signed* area — portions below the axis contribute negative amounts.



Exam tip — "area" versus "integral" can differ. If the curve dips below the x-axis on the interval of interest, the integral gives signed area (parts below count as negative) while the *geometric area* is always positive. For geometric area you must (i) find where f crosses the axis inside $[a, b]$, (ii) split the interval there, and (iii) take the absolute value of each piece before summing. Always sketch the curve first.

Worked Example 4 — Signed vs unsigned area

Compute $\int_0^3 (x^2 - 2x)\,dx$ and also find the total geometric area between $y = x^2 - 2x$ and the x-axis on $[0, 3]$.

```
Antiderivative: ∫ (x² − 2x) dx = x³/3 − x².
Signed integral:
   [x³/3 − x²]₀³ = (9 − 9) − (0 − 0) = 0.
The signed integral is ZERO — the positive and negative parts cancel.

Find the zero of the integrand inside [0, 3]:
   x² − 2x = x(x − 2) = 0  ⇒  x = 0 or x = 2.
On (0, 2) the curve is BELOW the axis (f(1) = −1 < 0).
On (2, 3) the curve is ABOVE the axis (f(2.5) = 1.25 > 0).

Split and take absolute values:
   ∫₀² (x² − 2x) dx = (8/3 − 4) − 0 = −4/3   ⇒  |−4/3| = 4/3.
   ∫₂³ (x² − 2x) dx = (9 − 9) − (8/3 − 4) = 0 − (−4/3) = 4/3.
Total geometric area = 4/3 + 4/3 = 8/3.

// The integral is 0 but the area is 8/3 — two different numbers from the same qu
```

6. Area Between Curves

To find the area enclosed between two curves $y = f(x)$ (top) and $y = g(x)$ (bottom) on $[a, b]$, integrate the difference:

$$\text{Area} = \int_a^b \big[f(x) - g(x)\big]\,dx, \quad \text{where } f \geq g \text{ on } [a, b].$$

The limits a, b are usually the x-coordinates of the intersection points, found by solving $f(x) = g(x)$. If the roles of "top" and "bottom" swap within the interval, split the integral at the crossover.

Worked Example 5 — Area enclosed by $y = x^2$ and $y = 4$

```
1.  Find intersection points: x² = 4 ⇒ x = ±2.
2.  On (−2, 2) the line y = 4 is ABOVE the parabola y = x².
3.     (top − bottom) = 4 − x².
4.  Area = ∫₋₂² (4 − x²) dx
5.       = [4x − x³/3]₋₂²
6.       = (8 − 8/3) − (−8 + 8/3)
7.       = (24/3 − 8/3) − (−24/3 + 8/3)
8.       = 16/3 − (−16/3)
9.       = 32/3.
10. Area enclosed = 32/3 square units.

// By symmetry of the region in the y-axis, 2 · ∫₀² (4 − x²) dx = 2 · 16/3 = 32/3
```

7. Integration by Substitution

Substitution is the integration analogue of the chain rule. If a composite function $f(g(x))$ has been multiplied by the derivative $g'(x)$ of its inner function, the substitution $u = g(x)$ collapses the integrand into a pure function of u.

Integration by substitution. If $u = g(x)$ is differentiable with $du = g'(x)\,dx$, then \[\int f(g(x))\, g'(x)\, dx = \int f(u)\, du. \] For a definite integral, the limits must also be transformed: when $x = a$, set $u = g(a)$; when $x = b$, set $u = g(b)$. Equivalently, you may substitute back to x before applying the original limits.

Recognising when substitution helps: the integrand should contain a function of $g(x)$ alongside (a constant multiple of) the derivative $g'(x)$. If you spot $g'(x)$ multiplying $f(g(x))$, substitute $u = g(x)$.

Worked Example 6 — Substitution with $u = x^2 + 1$

Find $\displaystyle\int \frac{2x}{x^2+1}\,dx$.

```
1.  Let u = x² + 1. Then du/dx = 2x, so du = 2x dx.
2.  The integrand contains 2x dx in the numerator — perfect.
3.  Rewrite:
4.     ∫ (2x / (x² + 1)) dx = ∫ (1/u) du.
5.  Integrate:
6.     ∫ (1/u) du = ln|u| + C.
7.  Substitute back:
8.     = ln|x² + 1| + C = ln(x² + 1) + C   (since x² + 1 > 0 always).

// Template: ∫ f'(x) / f(x) dx = ln|f(x)| + C — a very common pattern.
```

Worked Example 7 — Substitution with $u = x^2$

Find $\displaystyle\int 2x\,e^{x^2}\,dx$.

```
1.  Let u = x². Then du = 2x dx.
2.  Rewrite:
3.     ∫ 2x e^(x²) dx = ∫ e^u du.
4.  Integrate:
5.     ∫ e^u du = e^u + C.
6.  Back-substitute:
7.     = e^(x²) + C.
8.  Check by differentiation:
9.     d/dx [e^(x²)] = e^(x²) · 2x ✓.
```

Worked Example 8 — Linear substitution

Find $\int \cos(3x+1)\,dx$.

```
1.  Let u = 3x + 1. Then du = 3 dx, so dx = du/3.
2.  Rewrite:
3.     ∫ cos(3x + 1) dx = ∫ cos u · (1/3) du = (1/3) ∫ cos u du.
4.  Integrate:
5.     (1/3) sin u + C.
6.  Back-substitute:
7.     = (1/3) sin(3x + 1) + C.

// For any LINEAR inner function ax + b, the substitution just adds a factor of 1
```

Exam tip — for definite integrals, change the limits too. When you substitute $u = g(x)$ in $\int_a^b f(g(x))g'(x)\,dx$, the limits a, b are values of x. Under the new variable they become $g(a), g(b)$. Either (i) change the limits and never return to x, or (ii) keep the original limits and substitute back to x before evaluating. Mixing the two — evaluating u-limits on an x-expression — is a frequent and costly error.

8. Integration by Parts

Parts is the integration analogue of the product rule. Starting from $(uv)' = u'v + uv'$ and integrating both sides:

Integration by parts. If u, v are differentiable functions of x, then \[\int u\, \frac{dv}{dx}\, dx = uv - \int v\, \frac{du}{dx}\, dx, \] often written compactly as $\int u\,dv = uv - \int v\,du$. The technique trades one integral for another; for it to help, the new integral $\int v\,du$ must be simpler than the original.

Choosing u and dv: you want u to simplify on differentiation, and dv to remain integrable. A standard mnemonic is **LIATE**:

$$\text{Logarithmic} \prec \text{Inverse trig} \prec \text{Algebraic} \prec \text{Trig} \prec \text{Exponential}$$

Whichever type appears earlier in this list, set it as u. The remaining factor becomes dv.

Integrand type	Let $u =$	Let $dv =$	Reason
$x^n \ln x$	$\ln x$ (L)	$x^n\, dx$	$\ln x$ simplifies on differentiation.
$x^n e^{kx}$	x^n (A)	$e^{kx}\, dx$	x^n becomes a polynomial of lower degree.
$x^n \sin x$ or $x^n \cos x$	x^n (A)	trig dx	Polynomial degree decreases.
$\ln x$ (alone)	$\ln x$ (L)	dx	Differentiating $\ln x$ gives $1/x$.

Worked Example 9 — $\int x\, e^x\, dx$

```
LIATE: algebraic x before exponential e^x, so let u = x, dv = e^x dx.
Then du = dx and v = ∫ e^x dx = e^x.
Apply parts:
   ∫ x e^x dx = uv − ∫ v du
              = x e^x − ∫ e^x dx
              = x e^x − e^x + C
              = e^x (x − 1) + C.
Check: d/dx [e^x (x − 1)] = e^x (x − 1) + e^x = x e^x ✓.
```

Worked Example 10 — $\int x \cos x\, dx$

```
LIATE: algebraic before trig. u = x, dv = cos x dx.
du = dx;  v = sin x.
Parts:
   ∫ x cos x dx = x sin x − ∫ sin x dx
                = x sin x − (−cos x) + C
```

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                    = x sin x + cos x + C.
Check: d/dx [x sin x + cos x] = sin x + x cos x − sin x = x cos x ✓.
```

Worked Example 11 — $\int \ln x \, dx$ (the "trick")

```
No obvious product − but write ln x = (ln x) · 1.
Let u = ln x, dv = 1 dx.
du = (1/x) dx;  v = x.
Parts:
   ∫ ln x dx = x ln x − ∫ x · (1/x) dx
             = x ln x − ∫ 1 dx
             = x ln x − x + C
             = x (ln x − 1) + C.
Check: d/dx [x ln x − x] = ln x + 1 − 1 = ln x ✓.

// The "dv = dx" trick works for any single logarithm or inverse trig.
```

Worked Example 12 — $\int x^2 e^x \, dx$ (parts twice)

```
First pass. u = x², dv = e^x dx  ⇒  du = 2x dx, v = e^x.
   ∫ x² e^x dx = x² e^x − ∫ 2x e^x dx
               = x² e^x − 2 ∫ x e^x dx.
Use Worked Example 9: ∫ x e^x dx = e^x (x − 1) + C₁.
Substitute:
   ∫ x² e^x dx = x² e^x − 2 [ e^x (x − 1) ] + C
               = x² e^x − 2x e^x + 2 e^x + C
               = e^x (x² − 2x + 2) + C.
Check: d/dx [e^x (x² − 2x + 2)] = e^x (x² − 2x + 2) + e^x (2x − 2)
   = e^x [x² − 2x + 2 + 2x − 2] = x² e^x ✓.

// Rule of thumb: ∫ xⁿ e^(kx) dx needs parts applied n times.
```

Exam tip — LIATE is a guideline, not a law. It works in the vast majority of A-level integrals, but if LIATE gives an integrand that does not simplify, try the other way. The real test is whether $\int v\,du$ is simpler than $\int u\,dv$. Also, for integrands like $\int x \ln x\,dx$, LIATE sends you to $u = \ln x$ (correct), but always double-check by trying to integrate the alternative to see if it terminates.

9. Integration Using Partial Fractions

A rational function $P(x)/Q(x)$ (with $\deg P < \deg Q$) can often be decomposed into a sum of simpler fractions. Each piece integrates to a logarithm or power. For distinct linear factors in the denominator:

$$\frac{f(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}, \qquad \int \frac{1}{x-a}\,dx = \ln|x-a| + C.$$

Worked Example 13 — Partial fractions

Find $\displaystyle\int \frac{3x+5}{(x-1)(x+2)}\,dx.$

```
1.  Write: (3x + 5) / [(x − 1)(x + 2)] = A/(x − 1) + B/(x + 2).
2.  Multiply through:
3.     3x + 5 = A(x + 2) + B(x − 1).
4.  Cover-up at x = 1:   3(1) + 5 = A(3)   ⇒  8 = 3A  ⇒  A = 8/3.
5.  Cover-up at x = −2:  3(−2) + 5 = B(−3) ⇒ −1 = −3B ⇒  B = 1/3.
6.  Rewrite the integrand:
7.     (3x + 5) / [(x − 1)(x + 2)] = (8/3)/(x − 1) + (1/3)/(x + 2).
8.  Integrate term by term:
9.     ∫ ... dx = (8/3) ln|x − 1| + (1/3) ln|x + 2| + C.

// Always write the cover-up values at the "bad" points of the denominator.
// Combine the two logs if asked: = (1/3) ln|(x − 1)⁸ (x + 2)| + C.
```

10. Integration Using Trigonometric Identities

Many trigonometric integrands don't match the standard forms directly, but they can be rewritten using identities. The most useful identities for integration are:

Identity	Useful for
$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$	$\int \cos^2 x \, dx$
$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$	$\int \sin^2 x \, dx$
$\sin 2x = 2 \sin x \cos x$	$\int \sin x \cos x \, dx$
$\sec^2 x = 1 + \tan^2 x$	$\int \tan^2 x \, dx$
Product-to-sum, e.g. $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$	$\int \sin(mx)\cos(nx) \, dx$

Worked Example 14 — $\int \cos^2 x \, dx$

```
1.  Apply the double-angle identity cos²x = (1 + cos 2x)/2.
2.  Rewrite:
3.     ∫ cos²x dx = ∫ ½(1 + cos 2x) dx
4.                = ½ ∫ 1 dx + ½ ∫ cos 2x dx
5.                = ½ x + ½ · ½ sin 2x + C
6.                = x/2 + (sin 2x)/4 + C.
7.  Check: d/dx [x/2 + (sin 2x)/4] = 1/2 + (cos 2x)/2 = (1 + cos 2x)/2 = cos²x ✓.

// The same trick with sin²x = (1 − cos 2x)/2 gives ∫ sin²x dx = x/2 − (sin 2x)/4
```

Worked Example 15 — Product-to-sum

Find $\displaystyle\int \sin(2x)\cos(3x)\,dx$.

```
1.  Apply product-to-sum:
2.     sin A cos B = ½ [sin(A + B) + sin(A − B)].
3.     sin(2x) cos(3x) = ½ [sin(5x) + sin(−x)]
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                    = ½ sin(5x) − ½ sin x.        (sin(−x) = −sin x)
Integrate:
    ∫ sin(2x) cos(3x) dx = ∫ [½ sin(5x) − ½ sin x] dx
                         = ½ · (−1/5) cos(5x) − ½ · (−cos x) + C
                         = −(1/10) cos(5x) + ½ cos x + C.
```

Exam tip — cos²x and sin²x cannot be integrated directly. Students often try $\int \cos^2 x\,dx = \frac{\cos^3 x}{3}$ (applying the power rule as if $\cos x$ were x) — this is wrong, because differentiating $\cos^3 x/3$ gives $-\cos^2 x \sin x$, not $\cos^2 x$. The only A-level-safe method is to linearise with the double-angle identity first.

11. Volumes of Revolution

Rotating the region between a curve $y = f(x)$, the x-axis, and the lines $x = a$, $x = b$ through a full turn about the x-axis sweeps out a solid whose volume can be computed by integration. The thin-disk approximation: each thin strip of width dx at position x becomes a disk of radius $y = f(x)$ and thickness dx, contributing volume $\pi y^2\,dx$. Integrating gives:

Solid of revolution — volume formula. The solid obtained by rotating the curve $y = f(x)$ around the x-axis from $x = a$ to $x = b$ has volume \[V = \pi \int_a^b y^2\, dx = \pi \int_a^b \big[f(x)\big]^2\, dx. \] For rotation about the y-axis of the curve $x = g(y)$ from $y = c$ to $y = d$, \[V = \pi \int_c^d x^2\, dy = \pi \int_c^d \big[g(y)\big]^2\, dy. \]

Axis of revolution	Variable of integration	Volume formula
x-axis	x	$V = \pi \int_a^b y^2\,dx$
y-axis	y	$V = \pi \int_c^d x^2\,dy$

Worked Example 16 — Volume of revolution of $y = \sin x$ about the x-axis, $0 \le x \le \pi$

```
Apply the disk formula:
   V = π ∫₀^π y² dx = π ∫₀^π sin²x dx.
Use sin²x = (1 − cos 2x)/2:
   V = π ∫₀^π ½(1 − cos 2x) dx
     = (π/2) [x − (sin 2x)/2]₀^π.
Evaluate:
   at x = π:  π − (sin 2π)/2 = π − 0 = π.
   at x = 0:  0 − (sin 0)/2  = 0.
V = (π/2) · π = π²/2.

// Volume of the "lemon" obtained by rotating one arch of a sine about the x-axis
// Numerical: π²/2 ≈ 4.9348 cubic units.
```

12. First-Order Separable Differential Equations

A differential equation relates a function to its derivative(s). The simplest useful case is a *first-order separable* equation in which the derivative factors into a function of x times a function of y:

First-order separable differential equation. An equation of the form \[\frac{dy}{dx} = f(x)\, g(y) \] is called *separable*. Separate the variables by treating dy/dx as a quotient of differentials: \[\frac{1}{g(y)}\, dy = f(x)\, dx \quad\Longrightarrow\quad \int \frac{1}{g(y)}\, dy = \int f(x)\, dx + C. \] The resulting equation (after integrating both sides) is called the *general solution*. An *initial condition* $y(x_0) = y_0$ pins C down to give the *particular solution*.

Worked Example 17 — Solve $dy/dx = 2y$ with $y(0) = 3$

```
Separate variables (valid while y ≠ 0):
   (1/y) dy = 2 dx.
Integrate both sides:
   ∫ (1/y) dy = ∫ 2 dx
   ln|y| = 2x + C.
Exponentiate:
   |y| = e^(2x + C) = e^C · e^(2x).
   y = A · e^(2x)        (A = ±e^C, any nonzero constant.)
```

```
Apply the initial condition y(0) = 3:
   3 = A · e^0 = A  ⇒  A = 3.
Particular solution: y = 3 e^(2x).
Check: dy/dx = 6 e^(2x) = 2 · 3 e^(2x) = 2y ✓. y(0) = 3 ✓.

// dy/dx proportional to y gives exponential growth — the classic population /
// radioactive-decay model.
```

Worked Example 18 — Solve $dy/dx = xy$ with $y(0) = 1$

```
Separate:
   (1/y) dy = x dx.
Integrate:
   ∫ (1/y) dy = ∫ x dx
   ln|y| = x²/2 + C.
Exponentiate:
   y = A · e^(x²/2).
Initial condition y(0) = 1:
   1 = A · e^0  ⇒  A = 1.
Particular solution:  y = e^(x²/2).
Check: dy/dx = e^(x²/2) · x = x y ✓.
```

13. Applications in Mechanics (SUVAT Links)

Integration provides the link between acceleration, velocity, and displacement for a particle in motion along a line. If $a(t)$ is the acceleration as a function of time:

$$v(t) = \int a(t)\,dt + C_1, \qquad s(t) = \int v(t)\,dt + C_2.$$

The constants C_1, C_2 are fixed by initial conditions (usually $v(0) = u$ the initial velocity, and $s(0) = 0$ the starting position). For constant acceleration a, integrating gives the familiar SUVAT equations:

$$v = u + at, \qquad s = ut + \tfrac{1}{2}at^2.$$

The power of integration is that it handles variable acceleration too. If $a(t) = 6 - 2t$ and a particle starts from rest, then $v(t) = 6t - t^2$ and $s(t) = 3t^2 - t^3/3$ — no SUVAT formula applies, but integration still gives the answer immediately.

14. Applied Worked Examples

Worked Example 19 — $\int x \ln x \, dx$ (parts with LIATE)

```
LIATE: logarithmic before algebraic, so u = ln x, dv = x dx.
du = (1/x) dx;  v = x²/2.
Parts:
   ∫ x ln x dx = (x²/2) ln x − ∫ (x²/2)(1/x) dx
               = (x²/2) ln x − ∫ (x/2) dx
               = (x²/2) ln x − x²/4 + C
               = (x²/4)(2 ln x − 1) + C.
Check: d/dx [(x²/2) ln x − x²/4] = x ln x + x²/2 · (1/x) − x/2
                                 = x ln x + x/2 − x/2 = x ln x ✓.
```

15. Summary — When to Use Which Technique

Integrand pattern	Technique	Key formula
x^n or polynomial	Power rule	$\int x^n \, dx = x^{n+1}/(n+1) + C$
Standard trig / exp / log	Quote standard integral	See table in §3.
$f(g(x)) \cdot g'(x)$	Substitution $u = g(x)$	$\int f(u) \, du$

Product of different function types (algebraic × exp / trig / log)	Integration by parts, with LIATE	$\int u\,dv = uv - \int v\,du$
Rational function $P(x)/Q(x)$ ($\deg P < \deg Q$)	Partial fractions	Decompose, integrate each piece to $\ln$.
$\sin^2, \cos^2, \sin mx \cos nx$	Trig identity (linearise)	Double-angle or product-to-sum.
Area between f and g	Definite integral of difference	$\int_a^b (f-g)\,dx$
Volume of revolution about x-axis	Disk method	$V = \pi \int_a^b y^2\,dx$
$dy/dx = f(x)g(y)$	Separate and integrate	$\int dy/g(y) = \int f(x)\,dx + C$

Chapter-wide checklist for Integration.

- **Always write** $+C$ on every indefinite integral.
- **For definite integrals, no** C **— but always show** $[F(x)]_a^b$ and the subtraction $F(b) - F(a)$.
- **Substitution: change the limits** if it's a definite integral, or substitute back if indefinite.
- **Parts: use LIATE** to pick u. Check whether $\int v\,du$ is actually simpler.
- **Sign of area:** if the curve dips below the axis, split the integral at the zeros.
- **For area between curves:** find intersection points first, then (top − bottom).
- $\cos^2 x$ **and** $\sin^2 x$**:** always linearise with the double-angle formula before integrating.
- **Volumes of revolution:** it is y^2, not y — remember to square $f(x)$.
- **Differential equations:** after integrating, always apply the initial condition to fix C.
- **Check by differentiation** whenever time permits — a 10-second sanity check.

Typical OCR examiner complaints (from mark reports).

- Forgetting $+C$ in indefinite integrals — a reliable way to lose an easy mark.

- Changing the variable via substitution but forgetting to change the limits, or evaluating u-limits on an x-expression.
- Choosing u and dv the wrong way round in integration by parts and then getting stuck in a harder integral.
- Integrating $\cos^2 x$ as $\cos^3 x/3$ (wrong — not a power of x).
- In volumes of revolution, integrating y instead of y^2, or forgetting the factor of π.
- For "area between a curve and the x-axis", failing to check whether the curve crosses the axis — taking the integral alone when the answer required the geometric area.
- In partial fractions, cover-up arithmetic errors, or decomposing a non-proper rational function (where $\deg P \geq \deg Q$) without first doing polynomial division.
- Solving a differential equation but forgetting to apply the initial condition — leaving C unspecified when the question demanded the particular solution.

Formula snapshot (memorise).

- Power rule: $\int x^n\,dx = x^{n+1}/(n+1) + C$ (for $n \neq -1$). $\int 1/x\,dx = \ln|x| + C$.
- Trig: $\int \sin x\,dx = -\cos x + C$; $\int \cos x\,dx = \sin x + C$; $\int \sec^2 x\,dx = \tan x + C$.
- Exp: $\int e^x\,dx = e^x + C$; $\int e^{kx}\,dx = (1/k)e^{kx} + C$.
- Linear inside: $\int f(ax+b)\,dx = (1/a)F(ax+b) + C$.
- Definite: $\int_a^b f(x)\,dx = F(b) - F(a)$.
- Substitution: $u = g(x)$, $du = g'(x)\,dx$. Change limits for definite.
- Parts: $\int u\,dv = uv - \int v\,du$. Choose u by LIATE.
- Partial fractions: decompose, then each piece integrates to $\ln$ or a power.
- $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$; $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$.
- Volume of revolution: $V = \pi \int_a^b y^2\,dx$ (about x-axis); $V = \pi \int_c^d x^2\,dy$ (about y-axis).
- Separable ODE: $\int dy/g(y) = \int f(x)\,dx + C$; apply initial condition.